

4. Write a C program for polyalphabetic substitution cipher uses a separate monoalphabetic substitution cipher for each successive letter of plaintext, depending on a key.

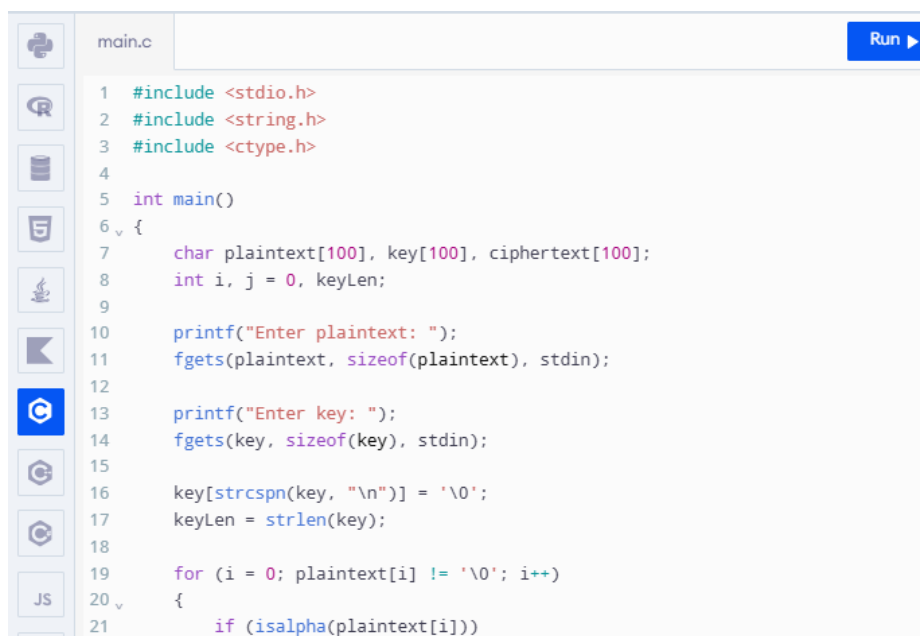
Aim:

To write a C program to implement a polyalphabetic substitution cipher, in which different substitution alphabets are used for successive letters of the plaintext based on a given key.

Algorithm

1. Start the program.
2. Read the plaintext from the user.
3. Read the encryption key.
4. Repeat the key until it matches the length of the plaintext.
5. For each plaintext character, calculate the corresponding ciphertext using:
$$C = (P + K) \bmod 26$$
6. Display the encrypted ciphertext.
7. Stop the program.

Program:



```
main.c Run ▶  
1 #include <stdio.h>  
2 #include <string.h>  
3 #include <ctype.h>  
4  
5 int main()  
6 {  
7     char plaintext[100], key[100], ciphertext[100];  
8     int i, j = 0, keyLen;  
9  
10    printf("Enter plaintext: ");  
11    fgets(plaintext, sizeof(plaintext), stdin);  
12  
13    printf("Enter key: ");  
14    fgets(key, sizeof(key), stdin);  
15  
16    key[strcspn(key, "\n")] = '\0';  
17    keyLen = strlen(key);  
18  
19    for (i = 0; plaintext[i] != '\0'; i++)  
20    {  
21        if (isalpha(plaintext[i]))
```

```

22     {
23         char p = toupper(plaintext[i]);
24         char k = toupper(key[j % keyLen]);
25
26         ciphertext[i] = ((p - 'A') + (k - 'A')) % 26 + 'A';
27         j++;
28     }
29     else
30     {
31         ciphertext[i] = plaintext[i];
32     }
33 }
34
35 ciphertext[i] = '\0';
36
37 printf("Ciphertext: %s", ciphertext);
38
39 return 0;
40 }

```

Output:

Output

```

Enter plaintext: HELLO
Enter key: KEY
Ciphertext: RIJVS

```

Result:

Thus, the polyalphabetic substitution cipher was successfully implemented in C using the given key to encrypt the plaintext.